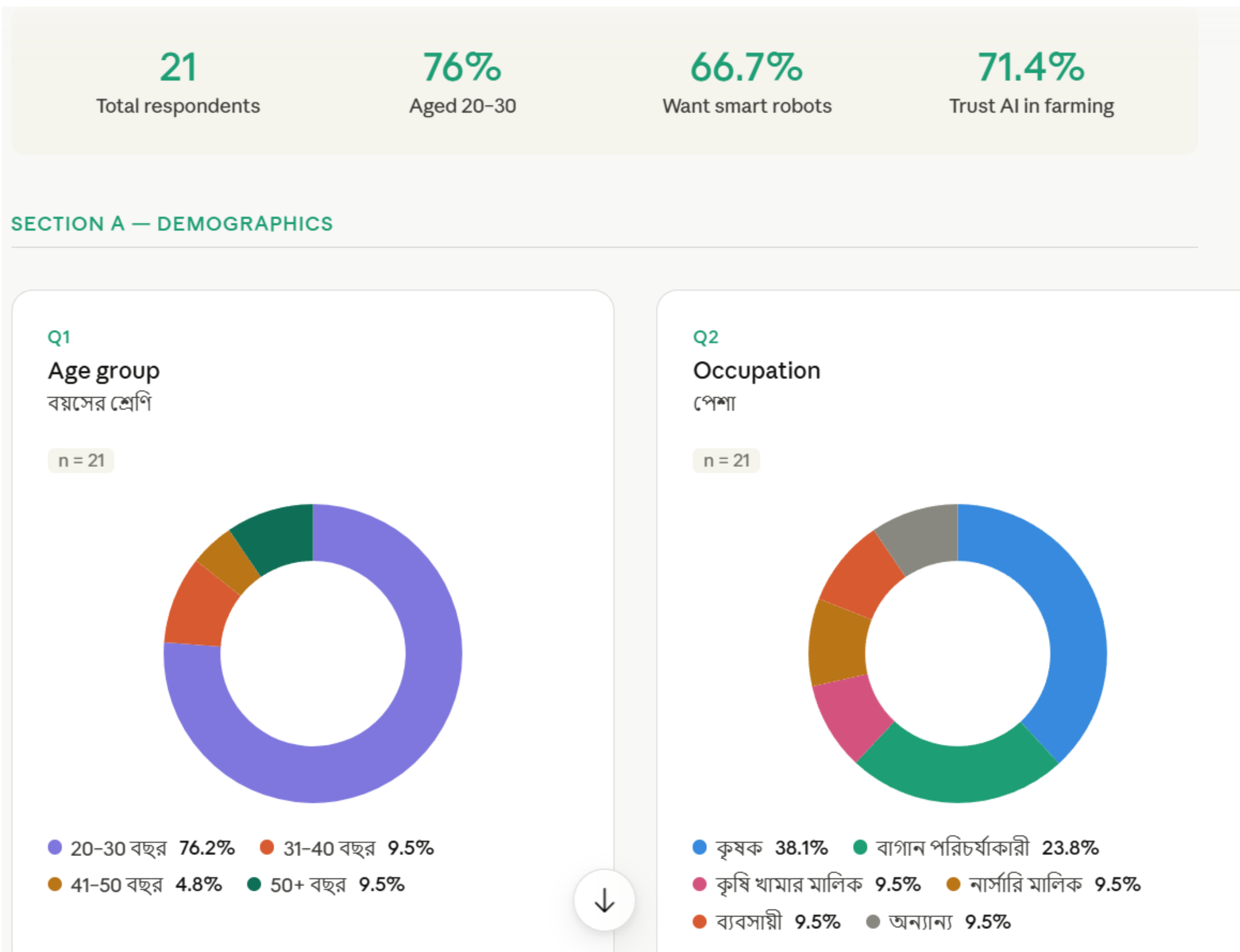
Independent University, Bangladesh

School of Engineering Technology and Science
Department of Computer Science and Engineering

CSE 426-PROJECT PROPOSAL FORM

SEMESTER: Autumn 2026
PROJECT TITLE: Smart Irrigation Rover for Adaptive Spot Watering

Survey to develop a process for complex engineering problems considering cultural and societal factors:



SECTION B — CURRENT IRRIGATION PROBLEMS

Q3

Involved in farming / plant cultivation?

কৃষিকাজ বা উদ্ভিদ চাষের সাথে জড়িত?

n = 21

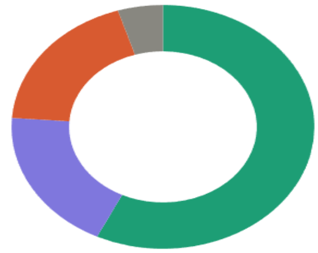
- না (No) 42.9%
- হ্যাঁ (Yes) 33.3%
- হতে পারে (Maybe) 23.8%

Q4

Does manual irrigation waste water?

হাতে পরিচালিত সেচে পানির অপচয় হয়?

n = 21



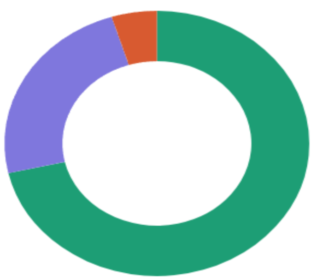
- একমত (Agree) 57.1%
- নিরপেক্ষ (Neutral) 19.0%
- দ্বিমত (Disagree) 19.0%
- উত্তর নেই 4.8%

Q5

Is manual irrigation time-consuming?

হাতে পরিচালিত সেচ কি সময়সাপেক্ষ?

n = 21



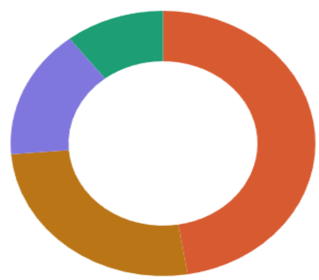
- একমত (Agree) 71.4%
- নিরপেক্ষ (Neutral) 23.8%
- দ্বিমত (Disagree) 4.8%

Q6

Difficulty monitoring soil conditions?

মাটির অবস্থা পর্যবেক্ষণে সমস্যা হয়?

n = 21



- হ্যাঁ (Yes) 42.9%
- কখনো কখনো (Sometimes) 23.8%
- কদাচিৎ (Rarely) 14.3%
- না (No) 9.5%

Q7

Can overwatering damage crops?

অতিরিক্ত পানি ফসলের ক্ষতি করতে পারে?

n = 21

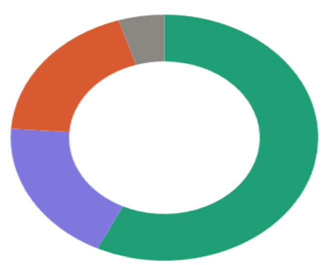
- হ্যাঁ (Yes) 57.1%
- কখনো কখনো (Sometimes) 23.8%
- না (No) 19.0%

Q8

Importance of soil moisture monitoring?

মাটির আর্দ্রতা পর্যবেক্ষণ কতটা গুরুত্বপূর্ণ?

n = 21



- অত্যন্ত গুরুত্বপূর্ণ 57.1%
- নিরপেক্ষ 19.0%
- গুরুত্বপূর্ণ নয় 19.0%
- উত্তর নেই 4.8%

SECTION C — TECHNOLOGY ACCEPTANCE

Q9

Automated irrigation robot reduces labor?

স্বয়ংক্রিয় সেচ রোবট শ্রম কমাতে সাহায্য করবে?

n = 21

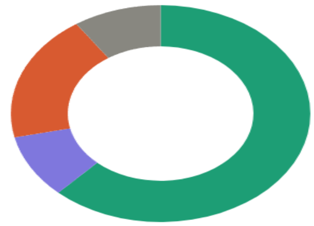
● সম্পূর্ণ একমত 9.5% ● একমত 57.1%
● নিরপেক্ষ 14.3% ● দ্বিমত 19.0%

Q10

Smart farming technology can improve agriculture?

স্মার্ট কৃষি প্রযুক্তি কৃষির উন্নতি করতে পারে?

n = 21



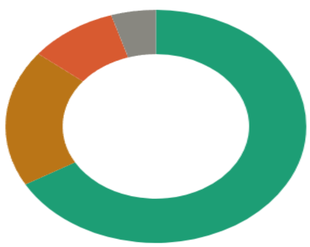
● একমত (Agree) 61.9% ● নিরপেক্ষ (Neutral) 9.5%
● দ্বিমত (Disagree) 19.0% ● উত্তর নেই 9.5%

Q11

Want auto dry-soil detection & irrigation?

স্বয়ংক্রিয়ভাবে শুকনো মাটি শনাক্ত করে সেচ চান?

n = 21



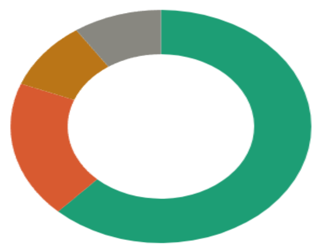
● হ্যাঁ (Yes) 66.7% ● হতে পারে (Maybe) 19.0%
● না (No) 9.5% ● নিশ্চিত নই 4.8%

Q12

Obstacle avoidance important for farming robots?

কৃষি রোবটের জন্য বাধা শনাক্তকরণ গুরুত্বপূর্ণ?

n = 21



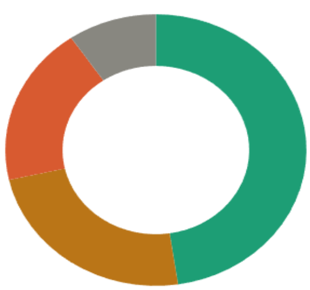
● হ্যাঁ (Yes) 61.9% ● না (No) 19.0%
● হতে পারে (Maybe) 9.5% ● নিশ্চিত নই 9.5%

Q13

Heard about IoT-based smart farming before?

IoT ভিত্তিক স্মার্ট কৃষি সম্পর্কে আগে শুনেছেন?

n = 21



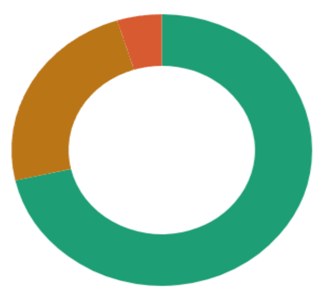
● হ্যাঁ (Yes) 47.6% ● হতে পারে (Maybe) 23.8%
● না (No) 19.0% ● নিশ্চিত নই 9.5%

Q14

AI & automation can improve future agriculture?

AI ও স্বয়ংক্রিয় প্রযুক্তি ভবিষ্যতের কৃষি উন্নত করবে?

n = 21



● হ্যাঁ (Yes) 71.4% ● হতে পারে (Maybe) 23.8%
● না (No) 4.8%

Q15

Interested in using low-cost smart farming robots?

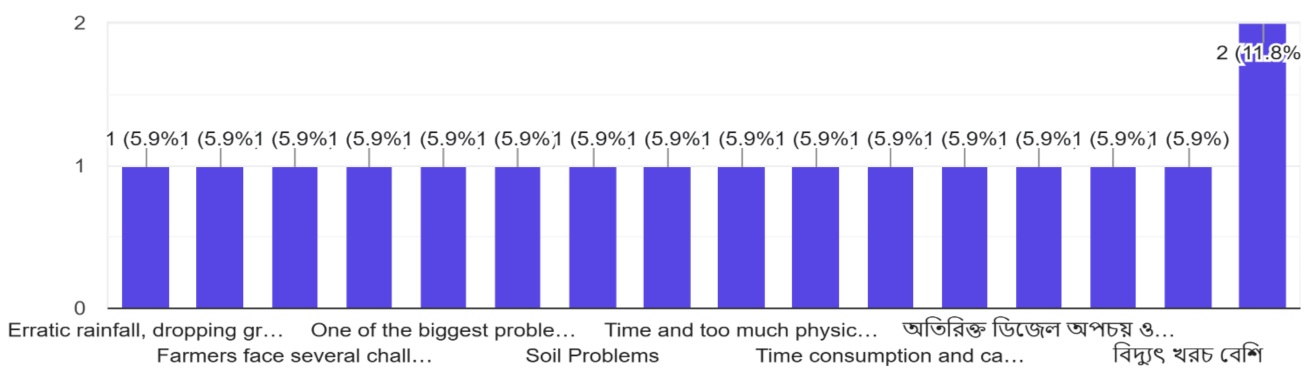
স্বল্পমূল্যের স্মার্ট কৃষি রোবট ব্যবহার করতে আগ্রহী?

n = 21

● হ্যাঁ (Yes) 66.7% ● হতে পারে (Maybe) 23.8% ● না (No) 9.5%

What are the biggest problems farmers face during irrigation? (সেচ প্রদানের সময় কৃষকদের সবচেয়ে বড় সমস্যাগুলো কী?)

17 responses



What features would you expect from a smart farming robot?(একটি স্মার্ট কৃষি রোবট থেকে আপনি কী কী সুবিধা বা বৈশিষ্ট্য প্রত্যাশা করেন?)

16 responses

crop monitoring, precision weeding, and harvesting using AI, sensors, and GPS navigation.

Automatic irrigation, crop monitoring, pest detection, and real-time farm data.

স্বল্প খরচে সেচ ব্যবস্থা, কম সময়ে আগাছা পরিষ্কার করা, ধান কাটা ও মাড়াই সম্পন্ন করা

A smart farming robot should include automatic irrigation, sensors for soil and weather monitoring, crop and pest detection, GPS navigation, remote control, and efficient water, fertilizer, and energy management to make farming easier and more productive.

Routine wise watering, detection of dehydration of plants, proper soil quality monitoring

Perfect irrigation system

সময়উপোযোগি সকল কাজ অতি দ্রুত ও কম খরচের রোবট প্রত্যাশা করি

রোগ- বালাই নিণর্য়, ও পরিচ্ছা

GOALS AND BENEFITS OF THE PROJECT:

GOALS OF THE PROJECT:

The main goal of this project is to develop a Smart Irrigation Rover for Adaptive Spot Watering that can move through a small agricultural area, measure soil moisture at different points, and apply water only to the dry spots. Instead of using a fixed soil moisture sensor or watering the whole area, the rover will identify moisture conditions at multiple soil points and perform spot-based irrigation.

Another goal is to classify soil conditions into different moisture zones such as Dry, Moderate, and Wet. Based on this classification, the rover will control the water pump for different durations so that very dry soil receives more water, moderately dry soil receives less water, and wet soil receives no water.

The project also aims to include post-irrigation verification, where the rover checks the soil moisture again after watering to confirm whether the irrigation was effective. In addition, IoT monitoring will be used to display soil moisture, temperature, humidity, zone status, and pump activity.

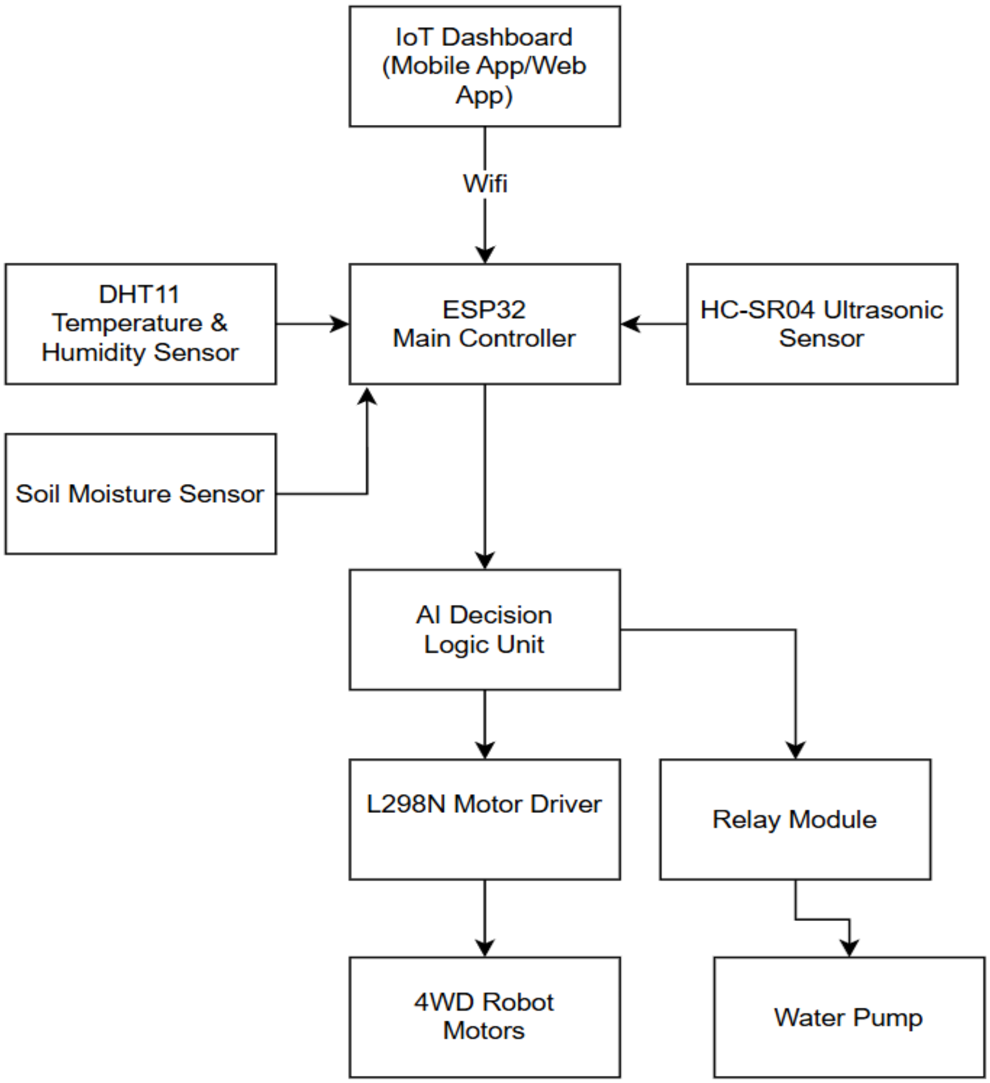
BENEFITS OF THE PROJECT:

This project can reduce unnecessary water usage by applying water only to the soil points that require irrigation. Since the rover checks multiple soil locations, it can identify dry spots more effectively than a fixed single-sensor irrigation system.

The adaptive spot watering method helps improve irrigation efficiency by controlling pump operation based on soil dryness level. The post-irrigation verification feature also helps confirm whether the watering action improved the soil moisture condition.

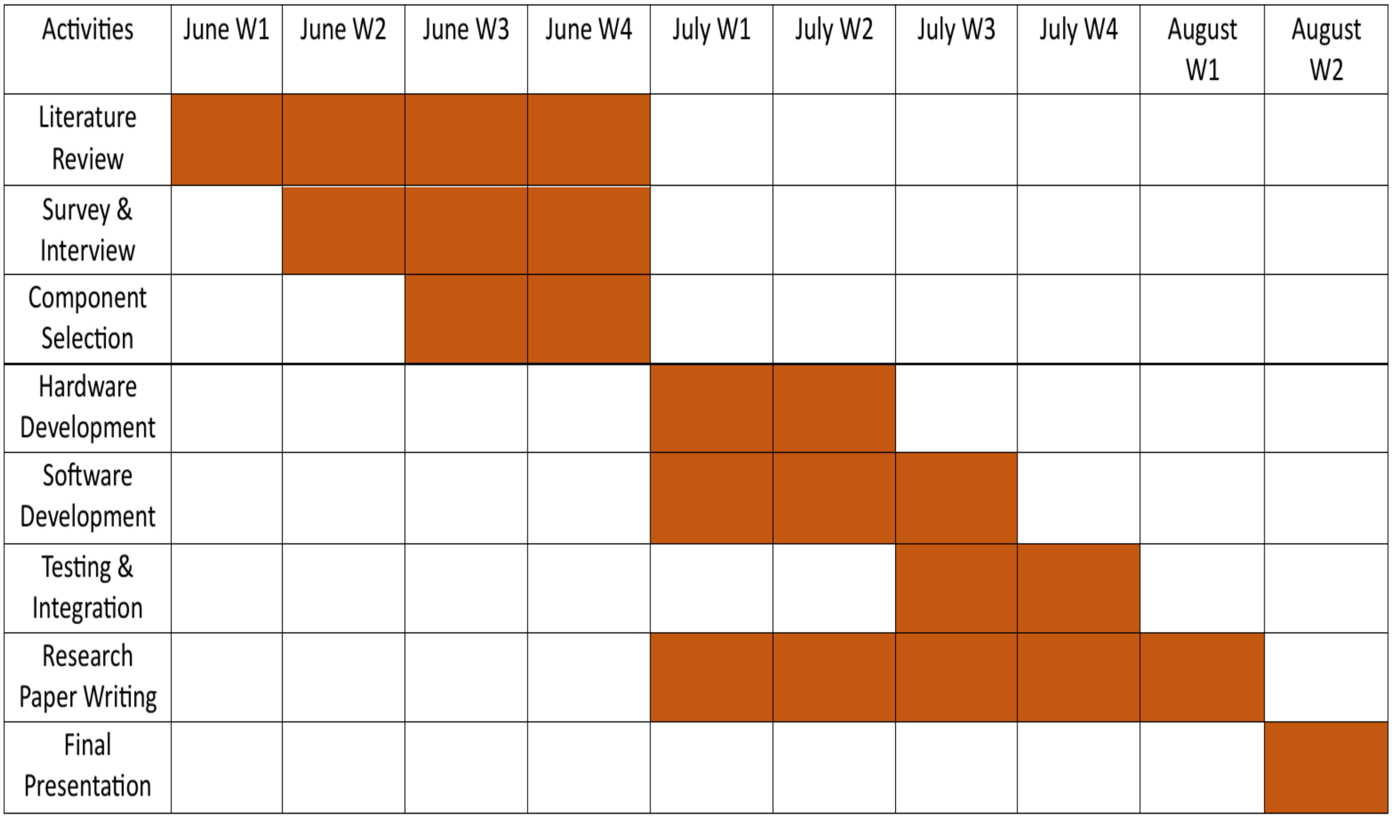
The system reduces manual checking of soil moisture and provides remote monitoring through IoT. Overall, the project demonstrates a low-cost smart irrigation solution that can support future research in precision irrigation, mobile soil monitoring, and autonomous agricultural robotics.

EXPERIMENTAL BLOCK DIAGRAM:



The proposed Smart Irrigation Rover for Adaptive Spot Watering is controlled by an ESP32 microcontroller, which acts as the central processing unit of the system. The soil moisture sensor, DHT11 sensor, and ultrasonic sensor continuously collect data about soil condition, environmental parameters, and obstacles. Based on the collected information, the system decides whether irrigation is required and controls the rover's movement and water pump accordingly. The rover can automatically move through the field, identify dry areas, and provide spot-based watering where needed. All important data, including soil moisture, temperature, humidity, and irrigation status, are transmitted to an IoT dashboard for real-time monitoring and analysis.

PROJECT TIMELINE (GANTT CHART):



REFERENCES:

[1] K. Obaideen, B. A. A. Yousef, M. N. AlMallahi, Y. C. Tan, M. Mahmoud, H. Jaber, and M. Ramadan, “An overview of smart irrigation systems using IoT,” Energy Nexus, vol. 7, 2022.

[2] M. Ayaz, M. Ammad-Uddin, Z. Sharif, A. Mansour, and E.-H. M. Aggoune, “IoT-enabled smart agriculture: Architecture, applications, and challenges,” Applied Sciences, vol. 12, no. 7, 2022.

[3] V. Kumar S., C. D. Singh, K. V. R. Rao, M. Kumar, Y. A. Rajwade, B. Babu, and K. Singh, “Evaluation of IoT based smart drip irrigation and ETc based system for sweet corn,” Smart Agricultural Technology, vol. 5, 2023.

[4] M. M. Alboaneen et al., “IoT-enabled smart agriculture for improving water management: A smart irrigation control using embedded systems and Server-Sent Events,” Results in Engineering, vol. 25, 2025.

[5] C. Han, W. Wu, X. Luo, and J. Li, “Visual navigation and obstacle avoidance control for agricultural robots via LiDAR and camera,” Remote Sensing, vol. 15, no. 22, 2023.

[6] F. Soto-Meca, J. M. Molina-Martínez, Á. Soto-García et al., “Comparative analysis of soil moisture- and weather-based irrigation scheduling for drip-irrigated lettuce using low-cost IoT capacitive sensors,” Sensors, vol. 25, no. 5, 2025.

[7] Z. Chen et al., “A comprehensive review of obstacle avoidance for autonomous agricultural machinery in multi-operational environment,” Artificial Intelligence in Agriculture, 2025.

[8] S. Mondal, A. M. Adar, M. Shidujaman, and M. S. Munir, “Low cost AI-driven autonomous rescue bot for water-based life-saving missions,” in Proc. IEEE Conference, 2025.

[9] A. Saha, L. Ali, R. Rahman et al., “IoT based fruit quality inspection and lifespan detection system,” in Proc. IEEE ICCUBEA, 2024.

[10] I. D. Niloy, S. M. Hoque, M. Hoque et al., “Smart floor cleaning robot,” in Proc. IEEE TENSYMP, 2023.

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DATE

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